

Jingmin Wang

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EDUCATION BACKGROUND

University of Toronto, Toronto, ON 09/2026-05/2028
Master of Science in Applied Computing
University of Toronto, Toronto, ON 09/2021-05/2026
Bachelor of Applied Science in Computer Engineering (Minor in AI)
Cumulative GPA: 3.95/4.0

WORK EXPERIENCES

Seiko Epson Corporation, Toronto, ON 05/2024-05/2025
Machine Learning Research Scientist

- **Motion estimation (ECCV 2026):** Designed a novel and **cost-effective** optimization-based algorithm to estimate velocity and angular velocity of fast-moving spherical objects in **rolling shutter cameras**, achieving **~15% lower mean absolute error** than commercially available methods ([Paper PDF](#))
- Delivered a state-of-the-art **stereo-matching depth** model utilizing a cost-volume based convolutional neural network architecture, achieving **<2 mm error** on industrial parts.
- Improved model performance by **20%** with introducing a surface normal loss; cut training time from **days to 8 hours** using optimization tools including mixed-precision, gradient clipping, etc.
- Led fine-tuning experiments on OOD data, resulting in **>30% performance gain** and **coverage expansion** from gears to broad part families.

Jiying Technology Co. Ltd., Toronto, ON 01/2025-06/2025
Part-time Researcher (Remote)

- **Transformer-based drug discovery:** Developed a novel method combining **graph neural networks** and **transformers** for drug discovery; achieving over **95%** accuracy on internal benchmarks ([Paper PDF](#))
- Fixed critical architecture issues such as linear projection for residual paths and bugs in the transformer attention mechanism, improving model stability and enhancing results.
- Designed robustness experiments, such as injecting noise into the graph and varying the negative sample ratio, showing **minimal degradation** under structured noise injections to the graph

RESEARCH EXPERIENCES

Data Driven Decision Making Lab, Toronto, ON 06/2025-Present
Machine Learning Researcher

- **Offline reinforcement learning (ICML 2025):** Designed an epistemic-uncertainty aware offline planning algorithm that achieves **~12%** improvement over previous state of the art methods ([Paper PDF](#))
- **World Models (Target: ICML 2027):** Investigating the optimal way to train world models to facilitate downstream gradient based planning. Developed a **novel objective function form** with a theoretical regret bound
- Built an 1D Advanced Reasoning Corpus dataset benchmark with **200k+ data points** and standard framework with clean APIs and docs to compose function primitives at scale. Trained **transformer encoders and other LLMs** on these datapoints.
- Maintained a unified reinforcement learning codebase (**20+ algorithms, ~40k LOC**) while implementing seven new algorithms, refactoring modules, and drastically improving overall code coherence.

Learning, Embodied Autonomy, and Forecasting Lab, Toronto, ON 01/2025-Present
Reinforcement Learning Researcher

- **Model-based reinforcement learning (under consideration, ICLR 2027):** Designed a model-based preference-based RL algorithm that achieves **~15%** sample efficiency and **~10%** improved success rate than previous state of the art model-free algorithms. Developed a **first-of-its-kind** sublinear regret bound for our training objective ([Paper PDF](#)).
- Developed a **value-aware** method of learning a dynamics model from offline transitions to outperform maximum likelihood estimation. Designed objectives which apply the **value-aware model learning** framework to the offline setting.

- Developed novel **Q-function regularization objective** for offline RL to be used with maximum likelihood training with defining concepts such as ‘valid’ and ‘invalid’ actions for a continuous state and action space.

Intelligent Sensory Microsystems Lab, Toronto, ON

05/2023-09/2023

Computer Vision Researcher

- Engineered custom banded mask patterns to enhance light exposure control and reduce glare by filtering out indirect light.
- **Image sensing (IEEE Symposium on VLSI Technology and Circuits 2024)**: Participated in developing a software-defined imaging that intelligently allocates high-speed sensing in only in parts of the scene that need it ([Paper PDF](#))

TEACHING AND LEADERSHIP

Project Director and Academic Director in Reinforcement Learning, Machine Intelligence Student Team

University of Toronto, Toronto, ON

08/2022-05/2026

- Held annual **4-week lecture series and workshop** about reinforcement learning concepts covering the basics to policy gradient methods; taught **hundreds** of students in three years and received most entirely positive feedback (find a sample lecture here: <https://www.youtube.com/watch?v=WNIE8I9IFGo>)
- Organized the **first ever** reinforcement learning competition at UofT, and delivered a keynote address to hundreds of students
- Directed a project, leading **8-10 developers** to construct a novel framework based on reinforcement learning and natural language processing for real-time Forex and commodity trading.

PROJECTS

Robust Nash-Equilibrium Seeking in Decentralized Networks ([Project Poster](#))

April 2026

- Develop a first-of-its-kind Nash-equilibrium-seeking algorithm to mitigate adversarial communication in multi-agent systems, supporting time-varying and constrained networks.
- Created an adversarial filtering module based on the geometric median of incoming messages, reducing error to the Nash equilibrium by **>100000x** compared with baselines in high agent settings. Reduced adversarial corruption by **>100000x** compared with the baseline.
- Obtained a Certificate of Distinction at the 2026 Capstone Design Fair (top 15-20 teams)

PUBLICATIONS

- Reflect-then-Plan: Offline Model-Based Planning through a Doubly Bayesian Lens. International Conference on Machine Learning 2025
- Velocity and angular spin estimation for fast-moving spherical objects with a rolling shutter camera. European Conference on Computer Vision 2026
- Uncertainty Aware Planning for Preference-Based Reward Learning in Reinforcement Learning. Under consideration at International Conference on Learning Representations 2027
- Graph Network-Based Analysis of Disease-Gene-Drug Associations: Zero-Shot Disease-Drug Prediction and Analysis Strategies. bioRxiv 2024
- 23,000-Exposures/s 360fps-Readout Software-Defined Image Sensor with Motion-Adaptive Spatially Varying Imaging Speed. IEEE Symposium on VLSI Technology and Circuits 2024

AWARDS

International Student Scholarship, University of Toronto

07/09/2021-01/05/2025

University of Toronto Excellence Award, University of Toronto

10/05/2023 & 10/05/2025

Edward S. Rogers Sr. Admission Scholarships, University of Toronto

14/10/2021

Dean's Honour List, University of Toronto

6 semesters in a row

SKILLS

Programming Language & Tools: Python, C#, C++, ARM Assembly, C, Verilog, Visual Studio Code, Git, Glade

Libraries/Agentic Frameworks: PyTorch, NumPy, Codex, Unity, Unity ML Agents, OpenAI Gymnasium, Matplotlib, OpenCV, TensorFlow, Keras, GTK+, Pandas